

MOHAMED A. MOHAMED

Cloud Automation & Containerization Engineer

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PROFESSIONAL SUMMARY

Systems operations professional with six seasons of business-critical IT experience and hands-on AWS, Kubernetes, Docker, and Terraform engineering through independent projects. Builds repeatable infrastructure, secure delivery workflows, and operational automation, combining Windows systems administration with demonstrated Kubernetes upgrades, GitOps deployments, and automated database recovery.

TECHNICAL SKILLS

Cloud Platforms: AWS, EKS, EC2, VPC, S3, ECR, IAM, IRSA, CloudTrail, Application Load Balancer

Containerization & Orchestration: Docker, Kubernetes, Helm, Kustomize, Argo CD, Argo Rollouts, Kubernetes Operators

Automation & IaC: Terraform, GitHub Actions, OIDC, CI/CD, GitOps, Kyverno, Conftest, Trivy

Scripting & Operating Systems: Python, Bash, PowerShell, Go, Linux, Windows Server, Active Directory, Group Policy

Networking/Methodologies: TCP/IP, DNS, subnetting, routing, load balancing, NetworkPolicies, least privilege, incident response, backup and recovery, Prometheus, Grafana

PROJECT PORTFOLIO

Independent engineering projects | 2026

Multi-AZ EKS Infrastructure & Cluster Lifecycle Management

Repeatable AWS environment for validating cluster upgrades, workload resilience, and least-privilege access.

Tech Stack: AWS EKS | Terraform | Bottlerocket | ALB | IRSA | Prometheus | Grafana

- Provisioned EKS infrastructure across 2 Availability Zones and 3 managed node groups using Terraform, then executed staged Kubernetes 1.35-to-1.36 upgrades with 99.966% observed HTTP request success across 26,611 test requests.
- Strengthened workload resilience with ALB readiness gates, spare capacity, and PodDisruptionBudgets, recording zero failures across 4,030 requests during a separate controlled node-drain test.
- Restricted workload access through prefix-scoped IRSA policies, validating 3 denied unauthorized S3 operations while preserving intended reads and listing, with CloudTrail evidence.

ForgePath | Secure Developer Platform

Local platform reference automating service creation, container validation, GitOps delivery, and controlled recovery.

Tech Stack: Backstage | Docker | Kubernetes | Argo CD | Argo Rollouts | Kyverno | Prometheus

github.com/SecureCloudOps/ForgePath

- Built centralized Backstage templates that bootstrap FastAPI services with 9 security controls and generate Kubernetes manifests from a single authenticated developer request.
- Engineered a delivery workflow that validates source and policies, builds signed container artifacts with SBOMs and provenance, and promotes immutable image digests through Git to Argo CD.
- Contained a defective canary at 5% traffic exposure and demonstrated 375-second recovery during a local incident exercise using Prometheus alerts, Argo Rollouts, and human-approved Git recovery.

KubeBackup | Kubernetes Database Backup Operator

Declarative PostgreSQL backup and restore automation with object storage and integrity verification.

Tech Stack: Go | Kubebuilder | Kubernetes | Docker | Terraform | S3/MinIO | GitHub Actions

github.com/SecureCloudOps/database-backup-operator

- Built a Go operator with 2 custom resources that reconcile PostgreSQL backup and restore requests into Kubernetes Jobs, upload backups to S3 or MinIO, and verify SHA-256 integrity before restoration.
- Validated controller behavior and recovery workflows with 21 passing controller specifications and 3 Kind end-to-end tests, checking backup persistence, restored row counts, and exact records.
- Restricted operator and backup-job access through least-privilege RBAC, default-deny NetworkPolicies, and non-root execution, and automated Cosign image signing through GitHub Actions.

PROFESSIONAL EXPERIENCE

Galoof & Associates | Roxbury, Boston, MA

System Ops Engineer | January 2021-Present | Seasonal, six tax seasons

- Administered 3 servers and 11 Windows endpoints supporting 9 users, maintaining tax applications, network connectivity, and document-processing systems throughout seasonal operations.
- Automated recurring monitoring, self-healing, maintenance, and configuration checks with Python and PowerShell to reduce repetitive administration and standardize system upkeep.
- Managed user identity and access through Active Directory accounts, permissions, and Group Policy, supporting controlled access to business systems.
- Standardized operations through 5 checklists and weekly backup validation, making maintenance and recovery preparation repeatable.
- Provisioned approximately 8 workstations and resolved application, email, networking, and peripheral incidents to keep tax preparation and filing workflows operational.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate

AWS Certified SysOps Administrator - Associate

Certified Kubernetes Administrator (CKA)

CompTIA Security+